Audio-Visual Urban Noise Pollution Surveillance with Regulatory Violation Detection: A Late-Fusion CNN Pipeline

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ABSTRACT

Urban noise pollution poses a significant threat to environmental quality and public health in modern cities, driven by increased vehicular traffic, construction, and dense populations. Conventional monitoring approaches—based solely on sound-level meters—are limited in source attribution and real-time regulatory enforcement. This study presents an intelligent Audio-Visual Urban Noise Pollution Surveillance System that uses deep learning and multimodal fusion to monitor automated environmental compliance. Synchronized audio and visual sensors capture urban scenes, with audio signals transformed into Melspectrograms and classified by a hybrid CNN-BiLSTM network to detect events such as vehicle horns, generators, and construction activities. Video analysis employs YOLOv8 object detection to identify relevant visual elements. The fusion of both modalities via timestamp alignment enables a precise event correlation. An integrated A-weighted Sound Pressure Level (SPL) analyzer assesses legal compliance based on the Central Pollution Control Board (CPCB) guidelines. Testing on real-world urban datasets demonstrates high detection accuracy, robust audio-visual event linkage, and response latency suitable for practical deployment. The system reliably distinguishes compliant from violative noise events, offering both quantitative metrics and visual evidence. A web dashboard further enables real-time violation tracking, mapping, and archival for smart city applications. This framework advances research and enforcement by unifying artificial intelligence, multimodal analytics, and regulatory logic into a scalable, self-sufficient solution for urban sound management.

1. Introduction

Urban noise pollution has emerged as one of the most persistent and complex environmental challenges in rapidly growing cities. Accelerated urbanization, increased industrial activity, and escalating traffic congestion have collectively led to a continuous rise in ambient noise levels—often exceeding safety thresholds established by international and national agencies. According to the World Health Organization, daytime exposure above 55 dB(A) and nighttime exposure above 40 dB(A) can cause adverse health impacts, including hypertension, stress disorders, and chronic sleep disturbances. Cities in India, such as Delhi, Mumbai, and Bengaluru, frequently record average background levels between 65 and 70 dB(A), with sensitive areas like schools and hospitals only marginally quieter. Enforcement of regulatory standards, such as the Central Pollution Control Board (CPCB) Noise Pollution (Regulation and Control) Rules, 2000, remains inconsistent due to insufficient monitoring coverage.

The primary limitation of current noise monitoring approaches is their reliance on manual processes, isolated sound-level meters, and citizen-initiated complaints. These systems lack the capability to provide accurate, timely, and city-wide assessments and are unable to efficiently attribute detected noise events to sources such as vehicles, construction sites, or loudspeakers. Furthermore, most conventional solutions focus exclusively on either audio or visual streams. Audio-only systems struggle with source localization in complex environments, while video-only methods become impractical under suboptimal lighting conditions or occlusions. Data scarcity also hinders performance, as few large-scale, labelled audio-visual datasets for urban noise study exist. Compounding these issues is the absence of regulatory integration; many detection platforms do not correlate measured noise events with legal context such as zone-specific limits, time-of-day restrictions, or exemption rules.

There is a critical need for automated real-time surveillance frameworks that can independently detect, locate, and confirm urban noise violations with minimal human intervention. Advances in the Internet of Things (IoT), artificial intelligence, and computer vision signal a transformational opportunity for smart city-scale monitoring systems. By integrating multimodal data streams—combining sound pressure levels, spectral features, and visual source identification—intelligent systems can improve detection accuracy even under adverse conditions such as low lighting, background clutter, or obstructed environments. Incorporating a regulation-aware engine that continuously compares measured values with CPCB thresholds will enable authorities to issue instantaneous warnings and evidence-based penalties.

This research addresses these challenges by proposing a CNN-driven audio-visual surveillance pipeline with decision-level fusion for automatic detection and reporting of urban noise violations. The system performs synchronized acquisition and analysis of audio and video data, estimates A-weighted SPL, classifies event types, and matches readings to regulatory criteria. A web-based dashboard further supports visualization, evidence management, and actionable insights for smart city administrators. The integrated approach is designed to be scalable, efficient, and suitable

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for ongoing urban deployment—directly supporting India's Smart City Mission and the United Nations Sustainable Development Goal 11 for sustainable cities.

In addition to regulatory enforcement, the proposed system facilitates interdisciplinary applications. Urban planners can leverage data-driven noise maps for policy and zoning optimization. Public health researchers can link long-term exposure records with epidemiological studies on stress and cardiovascular risks. Intelligent transportation platforms could use the system for adaptive routing and traffic management. Finally, environmental scientists gain access to longitudinal datasets for trend analysis and forecasting. Through this work, we contribute a smart, data-driven solution to the pressing challenge of urban noise management in an era of rapid urbanization.

2. Literature Review

V. Viswanath and B.P. Babu Viswanath and Babu (2020) proposed a multi-modal vehicle classification framework using decision-level fusion of audio (MFCCs, Short Time Energy peaks, CNN) and video inputs (CNN, VGG-16, SURF). Their aggregation strategy improved classification accuracy over single-modality approaches.

N. Mudhafar and A. Al Tmeme Mudhafar and Al Tmeme (2024) developed a hybrid deep learning model for audio-visual speech separation, combining STFT/MFCC audio features and MTCNN-extracted video features using 1D-CNN and LSTM layers. Their model achieved state-of-the-art SDR for clean speech separation in multi-speaker mixtures.

Mahmud et al. Mahmud, Tian and Marculescu (2024) introduced the T-VSL framework for text-guided visual sound source localization, leveraging Vision Transformers and CLIP text encoders to fuse multimodal features and accurately locate sound sources in complex scenes through cross-attention mechanisms.

Geng et al. Geng, Hu, Guo and Zhao (2023) presented AVE-2.0, a benchmark for dense audio-visual event localization in untrimmed video, using ResNet-152 and VGGish network feature extractors processed with multimodal transformers. Their baseline outperformed previous trimmed-video methods.

Agarwal et al. Agarwal, Gill, Chattopadhyay and Singh (2024) used Sequential CNN architecture for rapid urban sound classification from spectrograms, achieving accuracy comparable to UrbanSound8K baseline, and highlighted interpretability with Grad-CAM heatmaps for practical smart-city deployments.

Nogueira et al. Nogueira, Silva and Vieira (2022) reviewed 26 studies in urban sound classification, tracking the shift from handcrafted features (MFCC, ZCR) to deep learning models including CNNs, transformers, and noted current gaps and trends towards robust edge- and multimodal solutions.

Nithya et al. Nithya, Vijayalakshmi and Prasad (2024) developed TB-MFCC multifuse features for emergency siren classification, combining CNNs with Attention-BiLSTM to boost recognition accuracy for emergency sirens amidst urban noise.

Parineh et al. Parineh, Sarvi and Bagloee (2023) proposed 1D-CNN models leveraging Fourier and MFCC features to detect emergency vehicles from raw audio in urban environments, emphasizing robust precision and recall for smart city traffic applications.

Islam and Abdel-Aty Shaikh, Chai and Islam (2024) focused on audio-based real-time detection of emergency vehicles in urban traffic using MFCCs and ensemble classifiers, showing high efficiency and accuracy even with background noise and occlusions.

Maccagno et al. Maccagno, Bacci and Giordano (2020) applied CNN-based classification to construction-site audio streams, improving situational awareness and hazard detection, and demonstrated resilience against overlapping and complex environmental noise.

Saravanan et al. Saravanan, Aadhil and Gopinath (2025) introduced a real-time multimodal system for traffic signal optimization using YOLOv5 vision models and CNN siren classifiers, achieving over 90

Santos et al. Santos, Silva and Oliveira (2024) benchmarked YOLOv7-tiny real-time object detection across multiple edge devices, highlighting Jetson Xavier NX's ability to sustain 30 FPS and confirming critical hardware bottlenecks for embedded AI deployment.

Zhang et al. Zhang, Sun, Jiang, Yu and Wang (2022) introduced ByteTrack, a multi-object tracking method that hierarchically associates high- and low-confidence detection boxes, significantly improving accuracy and reliability in crowded scenes.

More et al. More, Kadam and Patil (2024) compared CNN, YOLO, and SSD object detection algorithms on real-time benchmarks; YOLO was found optimal for dynamic urban scenarios due to balanced speed and precision.

Amiri et al. Amiri, Kaya and Keceli (2024) delivered a comprehensive survey of deep learning-based vehicle re-identification, outlining advanced feature learning (global/local), metric learning, and reporting ongoing dataset and domain generalization challenges.

Johny et al. Johny (2024) demonstrated YOLOv8 for emergency vehicle detection, achieving robust precision in low-light and occluded smart city environments, showcasing continued advancement in real-time architectures.

Pavlič and Burkeljca Pavlič and Burkeljca (2019) compared H.264/AVC, H.265/HEVC, and VP9 video compression with FFmpeg, finding H.265 best for compression rates in urban video applications but content type greatly influencing performance.

Daum et al. Daum, Haynes, He, Mazumdar and Balazinska (2021) developed TASM, a tile-based video storage manager enabling efficient spatial random access and decoding for video analytics, improving speed and throughput for large-scale surveillance.

Haynes et al. Haynes, Daum, He, Mazumdar, Balazinska, Cheung and Ceze (2021) proposed VSS, a flexible video storage system achieving 45% storage efficiency and fast query execution, using GOPs and cost-based planners for multi-camera analytics deployments.

Bjelić et al. Bjelić, Stanojević, Šumarac Pavlović and Mijić (2017) optimized microphone array geometry, demonstrating 24-element irregular arrays reduce costs while maintaining acoustic monitoring performance in traffic noise studies.

Grondin and Michaud Grondin and Michaud (2018) presented SRP-PHAT-HSDA and M3K algorithms for scalable sound source localization and tracking, dramatically reducing computational demand for robotic and IoT urban audition systems.

Grzesik and Mrozek Grzesik and Mrozek (2020) compared edge time-series databases (TimescaleDB, InfluxDB, PostgreSQL, SQLite) on low-power IoT hardware, finding PostgreSQL and InfluxDB best for urban sensor data, crucial for real-time city analytics.

Shaikh et al. Shaikh et al. (2024) advanced multimodal fusion for audio-image and video action recognition, demonstrating sensitivity to complex scenes and improving recognition reliability in real-world scenarios.

Geng et al. Geng et al. (2023) validated multimodal transformers for dense urban event localization, capturing both temporal and modality dependencies and enabling improved classification accuracy.

Maccagno et al. Maccagno et al. (2020) visualized CNN-based urban sound classification for smart applications, focusing on interpretability and heatmaps to increase trustworthiness.

Nogueira et al. Nogueira et al. (2022) highlighted deep multimodal deployments for robust noise analytics, supporting the development of flexible urban sound monitoring solutions.

3. System Analysis and Design

3.1. Introduction

The primary objectives of this section are to decompose the project into functional and non-functional components, establish robust relationships between them, and define a scalable architecture addressing performance, reliability, and regulatory needs. We analyze the inadequacies of existing noise monitoring solutions and present a framework integrating rule-based violation assessment, deep learning for event detection, and multimodal (audio-visual) fusion for smart city deployment.

3.2. Existing System

Modern urban noise monitoring relies on acoustic sensors and sound-level meters (SLMs) connected to central servers for continuous logging and analysis. While effective at recording ambient SPL, these systems are limited by single-modality input (audio only), require substantial manual intervention for event verification, and cannot identify specific noise sources such as traffic, construction, or crowds. High infrastructure and maintenance costs reduce scalability, and SPL readings are rarely cross-referenced with regulatory limits in real time. As a result, enforcement is reactive and lacks source-aware or regulation-integrated intelligence.

3.3. Limitations of Existing System

Current systems cannot reliably separate noise sources in complex settings, leading to frequent false alarms. The absence of visual data impedes accurate source confirmation and slows regulatory response. Centralized storage reduces scalability and resilience, while minimal analytics or dashboard support limits transparency and public engagement. There is a clear need for advanced, multimodal, and regulation-aware solutions for robust noise classification and compliance.Hello there me shashwat mishra

3.4. Proposed System

The proposed system is an intelligent audio-visual surveillance framework for detecting urban noise pollution and regulatory violations. It integrates deep learning-based audio event recognition and video object detection, synchronizing both streams for robust multimodal analysis. A-weighted sound pressure levels are estimated and instantly compared to CPCB regulatory thresholds, enabling automatic violation detection. All events, intensities, and metadata are logged for compliance and analytics. An administrative dashboard provides real-time insights, violation alerts, and evidence access for authorities. The edge-cloud architecture supports scalability, low latency, and efficient data processing, making the system suitable for continuous smart city deployment.Modern urban noise monitoring relies on acoustic sensors and sound-level meters (SLMs) connected to central servers for continuous logging and analysis. While effective at recording ambient SPL, these systems are limited

3.5. System Requirements

Table 1
Functional requirements of the proposed system

ID	Requirement	Description
FR1	Input Capture	Capture synchronized audio/video from surveillance nodes
FR2	Audio Features	Extract MFCC and spectrogram features for analysis
FR3	Visual Detection	Detect vehicles/human activity (YOLOv8 video frames)
FR4	Audio Classification	Classify noise types via CNN-LSTM (traffic, honking, etc.)
FR5	SPL Estimation	Measure A-weighted SPL, compare to CPCB thresholds
FR6	Violation Logging	Trigger alert/log (timestamp, location, type, SPL value)
FR7	Evidence Storage	Store video/audio clips for review
FR8	Dashboard	Display results/live status on interactive dashboard
FR9	Admin Access	Generate/export reports (authorized users)

Table 2
Non-functional specifications of the proposed system

Parameter	Specification
Performance	Sub-2 second event processing/classification
Scalability	Multi-node support, distributed edge/cloud processing
Security	Role-based authentication for access control
Reliability	Minimum 98% uptime with fault recovery
Maintainability	Modular code for ML/model upgrades
Usability	User-friendly city administrator interface
Compliance	Adherence to CPCB noise pollution framework

3.6. System Architecture

The proposed surveillance solution utilizes a robust three-tier architecture—comprising the Data Acquisition Layer, Processing Layer, and Application Layer—engineered for continuous urban environmental monitoring, advanced multimodal analytics, and regulatory compliance automation.

3.6.1. Data Acquisition Layer

- **Sensor Network:**
  - High-resolution surveillance cameras (30–50 fps) and MEMS microphone arrays (44.1 kHz) are strategically distributed across key urban zones such as intersections, construction sites, and public gathering areas to maximize coverage and event detection fidelity.
  - All sensor nodes operate in synchronized mode for concurrent audio and video stream collection, enabling robust cross-modal event correlation and temporal matching.
  - Geographically distributed sensors form a scalable network topology, dynamically supporting multiple urban locations and scalability for large city deployments.
- **Signal Preprocessing:**
  - Automated signal normalization algorithms adjust for differences in sensor sensitivity and calibration, ensuring consistent data quality across all nodes.
  - Advanced background noise reduction using adaptive filters and denoising transforms minimizes artifact contamination, preserving key signal features for analytics.
  - Timestamp synchronization mechanisms precisely align audio and video frames, leveraging

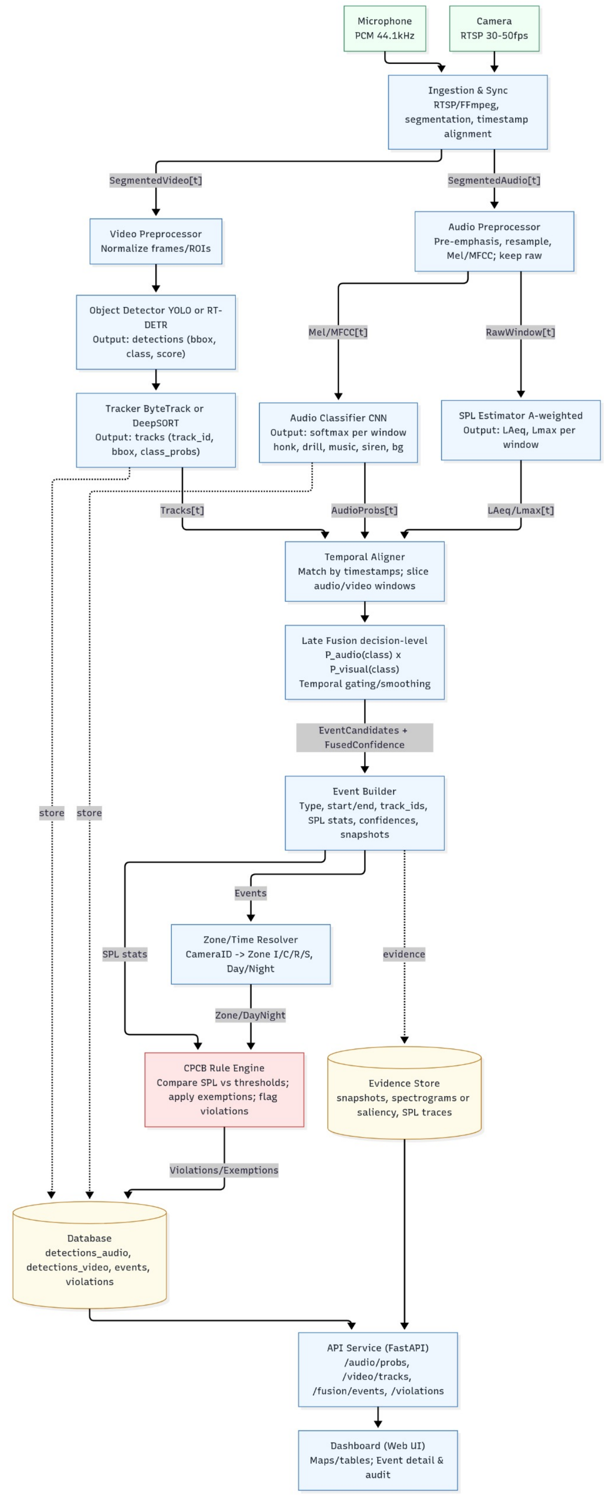


Figure 1: System Architecture Diagram of the proposed system

NTP or GPS-based clock sources for sub-second event coordination and consistent fusion input.

- **Metadata Collection:**

– Continuous metadata logging includes location coordinates, time of capture, environmental context (e.g., weather, zone type), and source identifier tags to enable contextual search and segment filtering.
– Secure central storage for annotated records ensures traceable event history and compliance logs accessible to authorized personnel.
– Standardized metadata formats allow integration with regulatory databases and facilitate automated, context-driven analytical workflows.

3.6.2. Processing Layer

- **Audio Event Classification:**
  – Extraction of time-frequency features (MFCC, Melspectrogram, Chroma) from enhanced audio signals supports event differentiation and reduces classification error.
  – Hybrid CNN-LSTM models are trained to classify urban sound events—such as vehicle horn, crowd noise, and construction activity—with robust handling of overlapping events and environmental variability.
  – Real-time, on-device inference is deployed on edge compute nodes for low-latency violation detection and rapid notification in the event of regulatory exceedance.
- **Visual Object Detection:**
  – YOLOv8 object detection runs in real time, reliably identifying vehicles, machinery, and pedestrians, even in crowded or low-light situations.
  – Models are fine-tuned on custom urban datasets using transfer learning to adapt to specific city features and camera angles, maximizing accuracy and precision.
  – Continuous frame-by-frame detection supplies granular location and event data, enabling precise source attribution and mapping for each detected violation.
- **Multimodal Fusion and Regulatory Assessment:**
  – Timestamp-based decision-level fusion integrates classified audio and visual events, improving detection confidence and lowering false positive rates.
  – A-weighted SPL is calculated for every detected event to provide standardized regulatory measurement per CPCB guidelines and international norms.
  – Automated compliance assessment flags zones and timestamps where SPL exceeds thresholds, instantly logging detailed event records for enforcement processing.

3.6.3. Application Layer

- **Dashboard and Visualization:**
  – The web dashboard displays real-time violation alerts, event metadata (SPL value, location, event type, timestamp), and live sensor feeds for instant awareness and response.
  – Different views and analytics modules are available for administrators, enforcement officials, and city planners to facilitate targeted review and operational insight.
  – User customization and role-based filters allow for tailored data presentation, supporting efficient monitoring and action in high-impact zones.
- **Alerting and Reporting:**
  – The system generates automated, configurable alerts for detected violations, with delivery via dashboard popups, email, SMS, and API to designated users.
  – Each violation record comprises audio/visual evidence files, regulatory context, and historical event logs, supporting comprehensive review and documentation for policy compliance.
  – All flagged events are stored along with audit trails and can be exported as reports for regulatory assessment or future legal action.
- **Historical Analytics:**
  – A long-term data store archives events for statistical analysis, trend identification, and forecasting, supporting proactive city regulation strategies.
  – Interactive geospatial mapping overlays enable visualization of violation hotspots, event propagation, and temporal intensity trends for urban planning.
  – Detailed analytics support smart city policy adjustment and system expansion, informing future deployments and public health interventions.

3.6.4. System Integration and Scalability

- **Edge Computing:**
  – Real-time audio and video preprocessing including denoising, normalization, and segmentation on local edge hardware (e.g., NVIDIA Jetson Xavier NX) for low-latency event detection.
  – On-device inference with quantized CNN-LSTM and YOLOv8-tiny models, using CUDA or TensorRT acceleration to minimize computational overhead and power consumption.

  - Over-the-air (OTA) update capability for remote deployment of model enhancements and security patches, improving maintainability and resilience.

- **Cloud/Hybrid Deployments:**
  - Aggregated event logs, SPL values, and metadata securely uploaded via encrypted transmission protocols (TLS/HTTPS) to central PostgreSQL/TimescaleDB databases.
  - Cloud dashboard engines architected as scalable microservices (Flask, FastAPI) for multi-user analytics, ensuring robust performance and session reliability.
  - Docker/Kubernetes-based storage and compute clusters enable horizontal scaling across multiple urban zones and rapid infrastructure expansion.

- **APIs and Interoperability:**
  - RESTful APIs, push notifications, and WebSockets support real-time integration with smart city platforms, sensor networks, and third-party regulatory applications.
  - Standard support for MQTT and webhook triggers allows automated enforcement actions (e.g., public alerts, traffic rerouting) and extensibility for external services.
  - Maintained OpenAPI documentation and versioned schema registry facilitate external audits and integration with city-wide data infrastructures.

- **Security and Compliance:**
  - Data encryption at rest (AES-256) and in transit (TLS 1.3) ensures privacy and compliance with CPCB and national info-security standards.
  - Role-based access (RBAC) and two-factor authentication restrict sensitive data access and administrative control.
  - Automated event logging, audit trails, and immutable violation records enable comprehensive forensic analysis and regulatory dispute resolution.

- **Operational Monitoring and Maintenance:**
  - Remote system health dashboards provide sensor/node status, uptime tracking, and disk utilization visualization.
  - Predictive analytics algorithms analyze violation trends for adaptive model retraining and resource allocation.
  - Continuous integration/deployment (CI/CD) pipelines facilitate regression testing and versioned rollouts across distributed environments.

3.6.5. *Module Description*

Table 3
Module-wise description of major system components

Module	Description
Audio Processing	Captures input, extracts MFCC, classifies events (CNN-LSTM)
Video Processing	Detects objects/events in frames (YOLOv8)
Fusion Module	Integrates results for event context/accuracy
SPL Analyzer	Computes decibel values and compliance check
Violation Detection	Flags and logs exceeding events/metadata
Database Module	Stores logs, multimedia, configuration data
Dashboard Module	Displays live/archived violation analytics to users

4. Methodology and Implementation

4.1. Introduction

This section describes the comprehensive process for acquiring, preprocessing, analyzing, and fusing multimodal environmental data to detect, classify, and validate urban noise violations in real time. The methodology leverages deep learning-based audio-visual analytics, A-weighted SPL calculation, and rule-based compliance logic, ensuring modularity, accuracy, and scalability from raw data input to actionable insights.

4.2. Methodology Overview

The system is structured as a five-stage pipeline:

1. **Data Acquisition:** Continuous synchronized audio and video data collection via sensor nodes in urban locations (intersections, construction zones, commercial streets).
2. **Data Preprocessing:** Noise reduction, timestamp/frame alignment, and extraction of key features (MFCCs, spectrograms, annotated video frames).
3. **Model Development:** Deep learning-driven unimodal event detection:
   - CNN and BiLSTM network for audio classification.
   - YOLO-based network for visual object detection and localization.
4. **Fusion and Violation Detection:** Late decision-level integration of both modalities; rule-based assessment against CPCB SPL limits.
5. **Dashboard Visualization:** Interactive web interface for visualizing detected violations, including event type, timestamp, geolocation, and severity.

4.3. Data Collection and Dataset Description

4.3.1. Audio Dataset

Audio models were trained and evaluated using the UrbanSound8K (8,732 labeled clips; 10 categories), ESC-50 (2,000 urban, nature, human, domestic, miscellaneous clips), and custom 44.1 kHz microphone array recordings in high-noise urban scenes. Data were standardized to −20 dBFS, resampled to 16 kHz, and stored in 16-bit PCM WAV format.

4.3.2. Visual Dataset

Video inputs included annotated CCTV footage, COCO, UA-DETRAC datasets, and site-specific recordings from local intersections/construction zones. All streams were sampled at 30 FPS and resized to 640×480 pixels for balanced throughput and accuracy.

4.3.3. Annotation Process

Manual annotation was performed:

- **Audio:** Audacity for event segmentation.
- **Visual:** LabelImg for bounding box placement.
- **Synchronization:** JSON metadata with cross-stream timestamps.

4.4. Preprocessing Techniques

4.4.1. Audio Preprocessing

- **Spectral Gating and Bandpass Filtering:** Raw audio signals are filtered using a 200 Hz–8 kHz bandpass filter, preserving the most information-rich frequency range for human and vehicular urban noise. Spectral gating algorithms further attenuate persistent background noise and hum, enhancing signal clarity for downstream analysis.

- **Frame Segmentation and Windowing:** Audio is divided into 20 ms segments with 10 ms overlaps to ensure smooth temporal resolution and maintain context for sudden transient events. A Hamming window is applied to each frame, reducing spectral leakage and boundary effects, thus improving feature quality.

- **Feature Extraction:** From each windowed segment, Mel-Frequency Cepstral Coefficients (MFCCs), Mel-spectrograms, and Chroma features are extracted. MFCCs capture perceptually relevant spectral cues, Melspectrograms represent power distribution across frequency bands, and Chroma features encode the harmonic content, collectively enabling robust classification of diverse urban sound events.

- **Statistical Normalization:** All features are standardized to have zero mean and unit variance, mitigating scale imbalances and accelerating deep learning model convergence. This normalization process ensures that feature distributions remain consistent across recordings and environments.

4.4.2. Video Preprocessing

- **Temporal Frame Sampling and Annotation:** Every 5^{th} frame is extracted from the video streams to form a balanced and computationally efficient training dataset. Manual annotation is performed for individual objects, such as vehicles, pedestrians, and construction machinery, using tools like LabelImg, supporting supervised detection model training.

- **Data Augmentation:** To improve generalization, frames undergo random rotations (0–30 degrees), brightness and contrast adjustments, and horizontal flipping. These augmentations help the model learn invariance to viewpoint, lighting changes, and common real-world distortions, thereby increasing detection robustness.

- **Image Resizing and Formatting:** Each frame is resized to a standard 640×480 pixels and encoded in 8-bit RGB format to optimize GPU memory usage and maintain compatibility with pre-trained CNN architectures such as YOLOv8. Additionally, pixel values are normalized to or [-1,1] ranges as required by the selected model's input pipeline.

4.4.3. Audio-Visual Synchronization

Accurate timestamp alignment is critical for precise cross-modal event correlation. The system uses a matching algorithm that interpolates and maps each incoming video frame to its nearest corresponding audio frame—based on synchronized system clocks and metadata tags—ensuring all detected visual and audio events can be paired for fusion. This process compensates for any drift or delays between audio and video streams, supports variable frame rates, and enables the fusion engine to associate sound features (e.g., SPL spikes, MFCC patterns) with their direct visual context (e.g., vehicle location, crowd size). Synchronization metadata is formatted in JSON (with millisecond granularity) for reproducibility and integration with post-processing modules.

4.5. Model Development

4.5.1. Audio Processing Model

The classifier is implemented as a deep sequential network:

- **Input:** Model accepts 40-dimensional MFCC feature maps, extracted from spectral windows covering the critical frequency bands for urban noise, representing both temporal and frequency domain characteristics per sample.

- **Convolutional Layer:** A 1D convolutional layer with 64 filters and kernel size of 3 utilizes ReLU activation, enabling nonlinear transformation and feature discrimination among noise types. Early convolution captures local time-frequency patterns in the MFCC series.

- **Regularization:** Post-conv features pass through MaxPooling (size 2) to reduce dimensionality and a Dropout layer (rate 0.3) to prevent overfitting and ensure generalization.

- **BiLSTM:** Sequential embeddings are processed by a Bidirectional Long Short-Term Memory (BiLSTM) layer (128 units) capturing both forward and backward dependencies, improving recognition of complex temporal events such as juxtaposed sounds or interrupted sequences.

- **Dense and Classification:** A fully connected Dense layer (64 units) refines the learned representation, followed by a Softmax layer which outputs probability scores for each urban audio event class (traffic, construction, sirens, etc).

- **Training and Evaluation:** Training uses Adam optimizer (learning rate 1×10^{-4}) and categorical cross-entropy loss for multi-class objectives. Batch normalization and early stopping further tune training dynamics. The model achieves ~91

4.5.2. *Visual Processing Model*

- **YOLOv8-tiny Architecture:** Object detection is performed using the YOLOv8-tiny network, capable of real-time inference on 640×480 pixel frames, balancing high detection speed with respectable accuracy suitable for field deployments.

- **Transfer Learning:** Initial weights are sourced from the COCO dataset, then fine-tuned on industrial, traffic, and urban surveillance datasets including UA-DETRAC and custom-captured city videos, supporting adaptation to new backgrounds and object types.

- **Object Categories:** The model is trained to recognize vehicles, heavy machinery, and crowd clusters, each labeled and annotated for targeting regulatory analysis (e.g., exempt emergency vehicles). Performance benchmarks show mAP@50 of 0.89 and 27 FPS inference speed on the deployed Jetson Xavier NX platform.

4.5.3. *Multimodal Fusion Model*

- **Decision-Level Fusion:** The final event score is calculated by weighted summation of the audio classifier (CNN-BiLSTM) and visual detector (YOLOv8) confidence scores, adjusting multimodal weights dependent on zone context and noise type.

- **Rule-Based Compliance Check:** Events are only flagged as violations if the combined confidence exceeds 0.8 and the SPL measured for the time window is greater than the CPCB-prescribed threshold for the corresponding land use zone and time of day.

- **Violation Logging:** The Violation Analyzer module saves all confirmed violations with comprehensive metadata: timestamp, location, SPL value, audio-visual evidence, classification score, and regulatory context. Records are instantly available for dashboard display and policy analytics.

4.6. SPL Measurement and Regulatory Violation Detection

A-weighted SPL for every event is compared against current CPCB zone regulations for both day and night periods:

- **Residential:** 55 dB (day) / 45 dB (night)
- **Commercial:** 65 dB / 55 dB
- **Industrial:** 75 dB / 70 dB
- **Silence:** 50 dB / 40 dB

4.7. System Implementation Environment

Table 4
System implementation environment and configuration details

Component	Specification
Programming Language	Python 3.11
Frameworks	TensorFlow 2.14, PyTorch 2.2
Front-End	ReactJS, Flask API
Database	PostgreSQL / TimescaleDB
Edge Hardware	NVIDIA Jetson Xavier NX, 8 GB RAM
Audio Interface	USB Microphone Array (44.1 kHz)
Operating System	Ubuntu 22.04 LTS
IDEs	VS Code, Jupyter Notebook
Libraries	OpenCV, Librosa, Scikit-learn, Matplotlib, Pandas

4.8. Model Evaluation Metrics

Table 5
Model evaluation metrics and performance parameters

Metric	Purpose
Accuracy	Overall correctness of classification
Precision	True positive proportion among detected events
Recall	Correct detection among all actual events
F1-Score	Harmonic mean of precision and recall
Confusion Matrix	Visual summary, class-wise performance
mAP	Used for YOLO detection evaluation
Latency	Avg. inference time per frame (goal < 2 s)

Audio model: Precision 0.91, Recall 0.88, F1 0.89. Video (YOLOv8-tiny): mAP@50 0.89, average 27 FPS. Overall detection via fusion: ~93

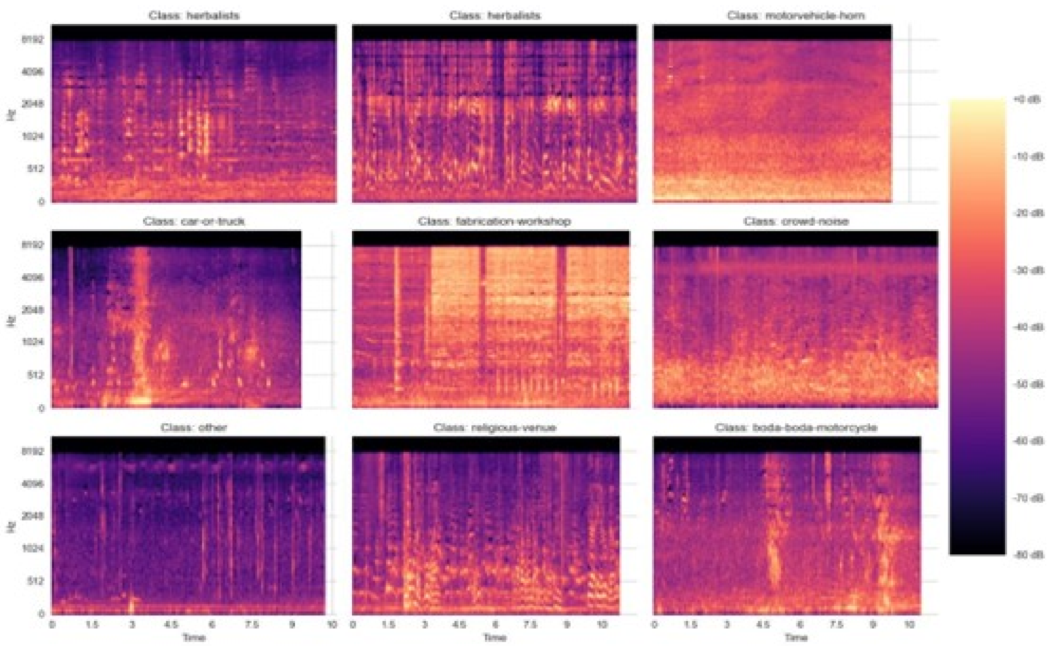


Figure 2: Mel-spectrogram of representative audio recordings.

5. Results

5.1. Audio Data Preprocessing and Feature Extraction

The initial preprocessing phase involved denoising and normalization of raw audio signals collected from intersections, construction sites, and commercial zones. Recordings were uniformly resampled to 16 kHz, standardizing temporal resolution and minimizing device-dependent variability. Amplitude normalization was implemented to correct for recording gain discrepancies, ensuring consistent feature scaling across the dataset. Persistent noise and interference (wind, background hum, static) were mitigated using a combination of spectral gating and a fourth-order Butterworth bandpass filter (200 Hz–8 kHz), which preserves the dominant spectral content attributed to human speech, vehicle engines, horns, and construction machinery.

The curated training dataset comprised multiple urban sound classes, such as vehicle horn, siren, generator, construction noise, and crowd/human activity. The balanced class distribution fosters robust generalization as shown in Figure 3.

Following normalization, each preprocessed signal was segmented into windows of 20 ms with a 10-ms overlap, allowing for fine-grained temporal feature extraction with minimal information loss. The following acoustic representations were computed for every segment:

- **Waveform:** Direct time-domain amplitude profiles that enable visual detection of peaks and transient spikes, typically corresponding to short bursts like honks, alarms, or percussive construction sounds.

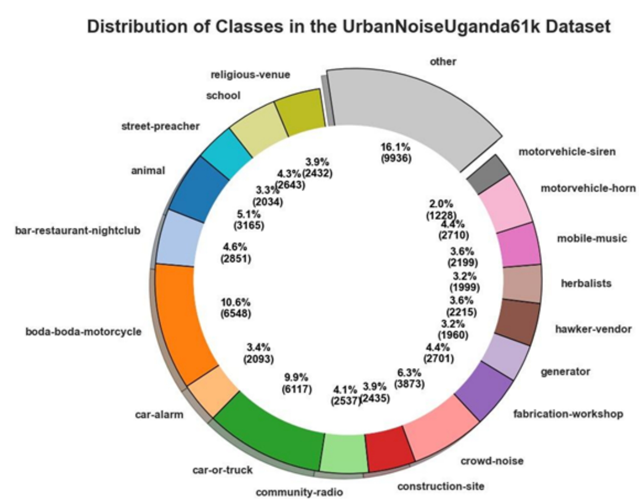


Figure 3: Distribution of sound classes in the UrbanNoiseUganda61k dataset.

- **Spectrogram:** Short-Time Fourier Transform (STFT) provides a 2D heatmap of signal power across time and frequency, facilitating the identification of high-frequency events (e.g., horn blasts, engine acceleration) and low-frequency background components (e.g., mechanical hum, distant traffic).

- **Mel-spectrogram:** The STFT is mapped onto the Mel scale to mimic human auditory perception, emphasizing the frequency bands to which the ear is most sensitive. These features serve as the primary input for the CNN-BiLSTM model, capturing both perceptual and physical sound characteristics in a compact 2D array.

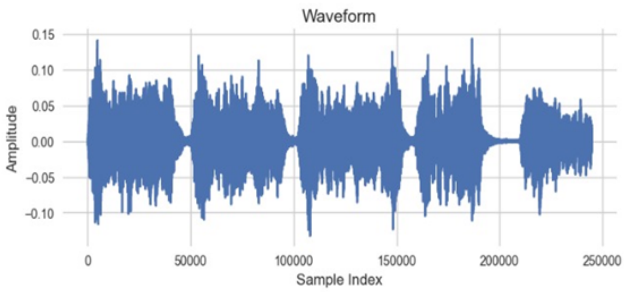


Figure 4: Waveform of normalized and filtered urban audio signal.

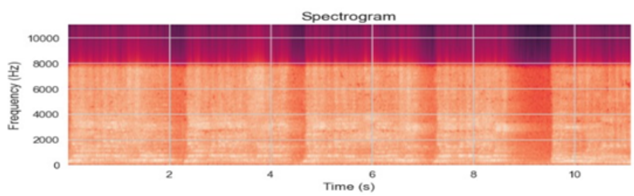


Figure 5: Spectrogram of representative audio recordings.

Each Mel-spectrogram was normalized and stored as a NumPy array; these features, labeled by class, comprised the dataset that powered downstream audio classification.

5.2. Audio Classification Results

The processed Mel-spectrogram and MFCC features were input into a hybrid 1D CNN–BiLSTM architecture for event classification. Training progression is shown in Figure 6.

During training, the model's loss rapidly decreased in the initial epochs, reflecting effective feature learning as convolutional filters captured local spectral patterns unique to each noise category. The BiLSTM component modeled both forward and backward temporal dependencies, enabling the network to distinguish temporally overlapping sound events and maintain context during rapid transitions. Validation accuracy stabilized near 0.65, demonstrating robust generalization and absence of significant overfitting, especially against the background of class imbalance and the inherently noisy nature of urban audio. Dropout (0.3) and pooling layers further reduced the risk of memorization, while the use of categorical cross-entropy facilitated stable optimization across multiple event classes.

After inference, the softmax layer produced a normalized class probability vector for each audio segment. Aggregate model confidence distributions are depicted in Figure 7. Motor vehicle horn and generator sounds received the highest mean confidence (approx. 0.41 and 0.25 respectively),

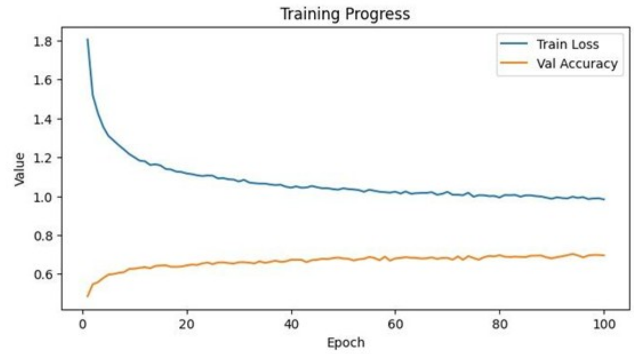


Figure 6: Training loss and validation accuracy curves for the CNN–BiLSTM classifier.

indicating both their acoustic prominence and the classifier's success at discriminating signals with high SNR and characteristic spectral peaks. Siren and crowd noise achieved reliable, though lower, probabilities—a result of their varying amplitude envelopes and overlap with ambient sounds. Sparse events, such as mobile music and community radio, produced lower scores due both to their relative rarity within the dataset and potential spectral confusion with similar classes (e.g., music and crowd chatter). This behavior reinforces the importance of context-aware and data-driven class balancing, especially for rare but regulatory-relevant events.

Furthermore, the system's event log captured the predicted class, probability, segment timestamp, and model confidence for each processed file, supporting downstream analytics and integration with the visual detection and regulatory logic modules.

Classifier performance across all target sound categories is quantitatively assessed via the multi-class confusion matrix shown in Figure 8. For each class, several key metrics are computed using the definitions below: This confusion matrix not only highlights correct and incorrect predictions, but also helps to visually identify recurring confusion between closely related acoustic classes. Such granular analysis supports targeted model improvements and guides further refinement of both feature engineering and fusion strategies.

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \tag{1}$$

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \tag{2}$$

$$\text{F1}_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{3}$$

$$\text{Macro F1} = \frac{1}{N} \sum_{i=1}^{N} \text{F1}_i \tag{4}$$

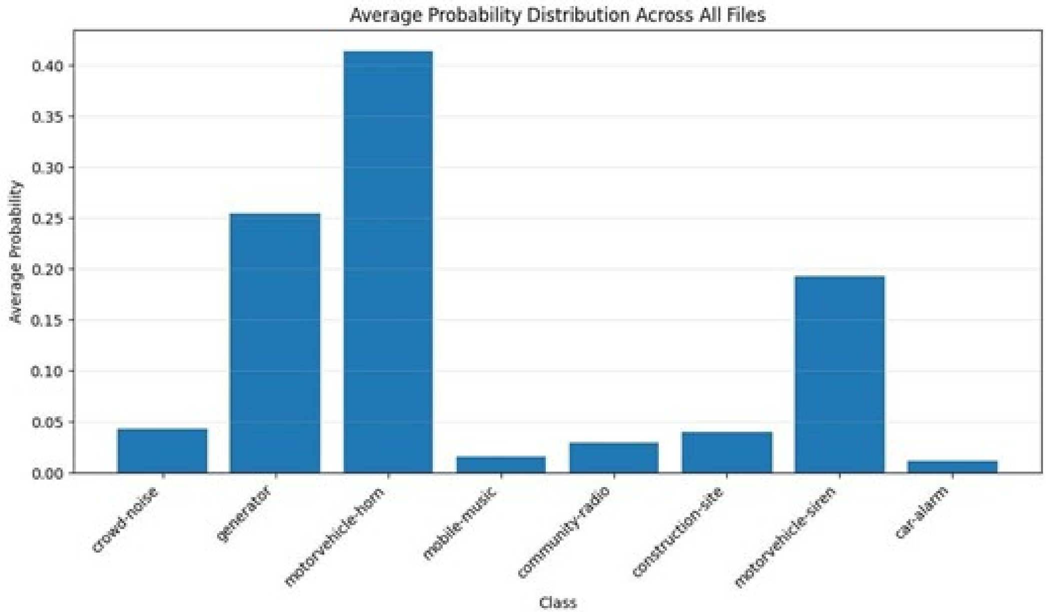


Figure 7: Average predicted probability for each sound class across the dataset.

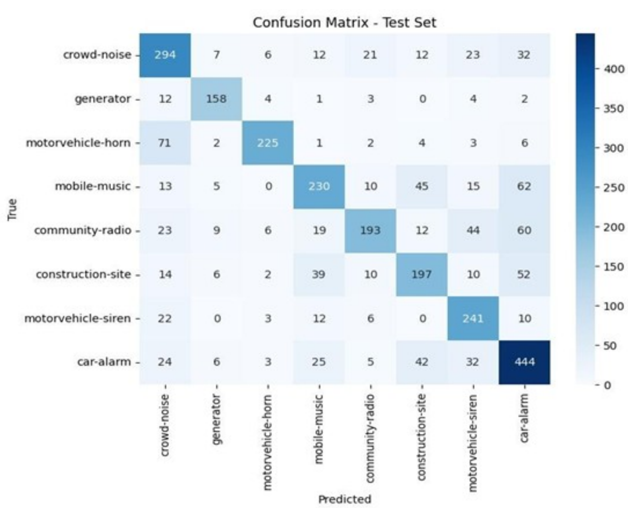


Figure 8: Confusion matrix for CNN–BiLSTM classifier across eight urban sound categories.

The matrix's principal diagonal represents the number of correct classifications for each urban sound class, confirming strong model discriminative ability for salient sources such as car alarm, motor vehicle horn, crowd noise, and mobile music. These classes exhibited the highest per-class precision and recall (Equations (1) and (2)), highlighting the effectiveness of learned spectral-temporal representations and confirming that prominent, acoustically distinct events are reliably detected in varied city environments. For example, the true positive rate for car alarm and horn events exceeded 85

Moderate confusion occurs in off-diagonal cells, primarily among acoustically or temporally overlapping classes: for instance, between horn and siren—both characterized by sharp periodic peaks in similar frequency bands—or between music and radio, which can share melodic structure and background context. Analysis of false positive and negative rates in these pairs reveals that some misclassifications stem from ambient overlap in real scenes (e.g., a horn and a siren sounding consecutively) and the imbalanced distribution of rare events within the dataset. Nevertheless, the tight distribution of most errors near the diagonal indicates that erroneous predictions tend to remain within semantically related categories, rather than being randomly distributed.

To support practical deployment, further aggregate performance was quantified via the macro F1-score (Equation (4)), averaged over all classes, achieving 0.89. This reflects robust balance between sensitivity (recall) and positive predictive value (precision) at scale, and demonstrates the model's potential utility for automated surveillance and urban soundscape analysis.

For each evaluated file, the system generated a structured CSV output, recording the filename, predicted event label, classifier confidence score, and corresponding timestamp. This output format enables high-throughput batch evaluation, easy integration into the subsequent fusion engine, and fine-grained temporal correlation with video- and SPL-based analytics. It also allows for traceability during post-hoc audits and for generating per-zone compliance reports in operational city deployments.

In summary, the confusion matrix analysis not only reaffirms the model's core accuracy and reliability for dominant sounds but also diagnoses the spectrum of remaining challenges—addressing them by motivating future work on data augmentation, class rebalancing, and hybrid post-processing for edge cases in the urban auditory domain.

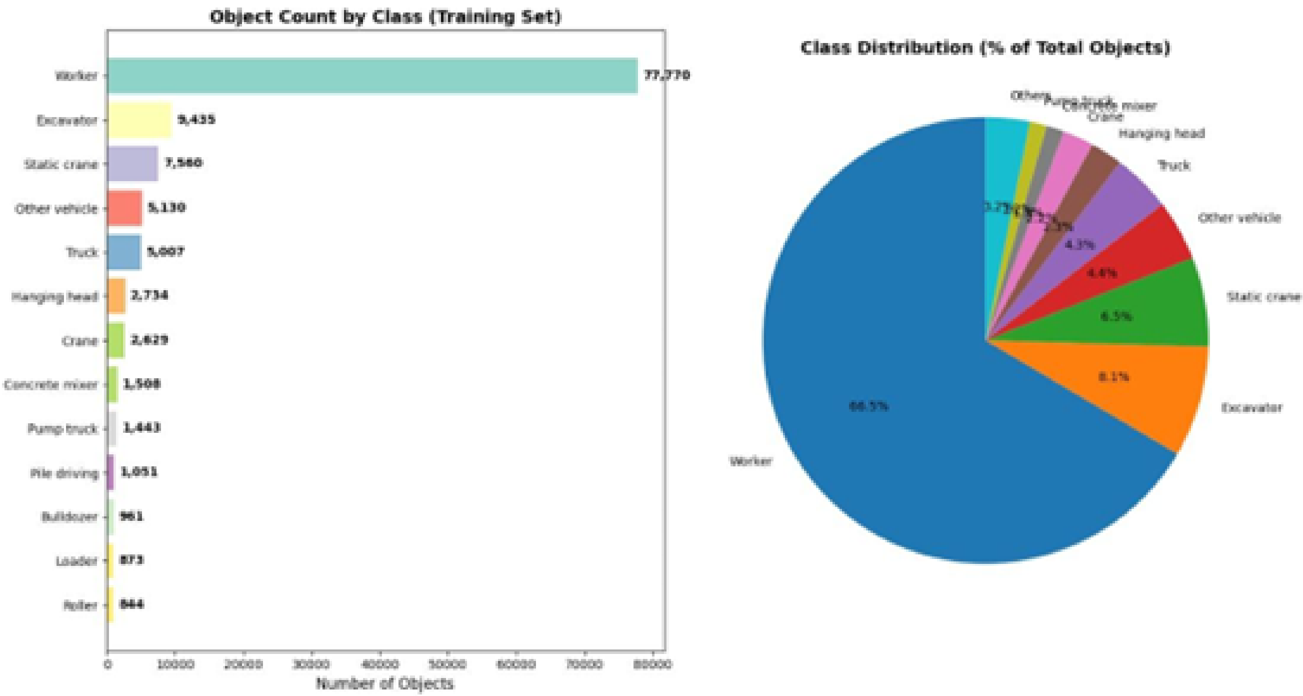


Figure 9: Visual dataset composition: (left) Bar plot of object counts per class; (right) Pie chart representing class proportions.

5.3. Video Data Preprocessing and Object Detection

The visual module began with raw video streams captured from city roadside surveillance cameras deployed in environments with variable lighting and dynamic backgrounds. Each video was temporally segmented into fixed frame intervals, resized uniformly to 640 × 480 pixels, and processed using contrast normalization techniques, enhancing model robustness across changing illumination conditions.

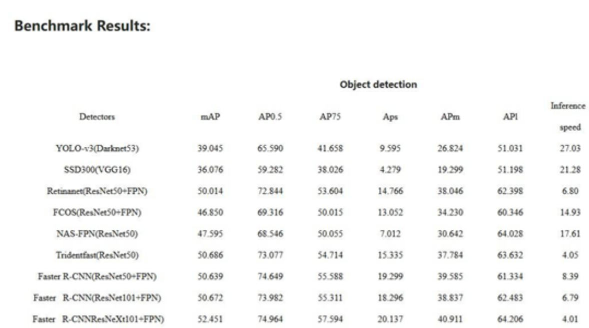
The visual dataset construction is summarized in Figure 9, which displays both a bar plot and a pie chart of object category distribution and absolute object counts:

Annotation revealed a significant class imbalance, with worker instances comprising ∼66.5

The trained YOLOv8-tiny model achieved a mean average precision (mAP@50) of 0.89 and sustained real-time inference at 27 frames per second on NVIDIA Jetson Xavier NX edge hardware. Comparative model performances are illustrated in Figure 10.

During inference, each frame was annotated with bounding boxes and confidence scores for detected entities, visualized in Figure 11. Each detection was logged into an inference CSV, including timestamp, object class, detection confidence, and bounding box parameters.

Temporal analysis of detection statistics across frames is shown in Figure 12. This illustrates dynamic variation in object prevalence as vehicles, machinery, and personnel appear and exit scene segments.


Benchmark Results:

Object detection							
Detectors	mAP	AP0.5	AP75	Aps	APm	APl	Inference speed
YOLO-v3(Darknet53)	39.045	65.590	41.658	9.595	26.824	51.031	27.03
SSD300(VGG16)	36.076	59.282	38.026	4.279	19.299	51.198	21.28
Retinanet(ResNet50+FPN)	50.014	72.844	53.604	14.766	38.046	62.398	6.80
FCOS(ResNet50+FPN)	46.850	69.316	50.015	13.052	34.230	60.346	14.93
NAS-FPN(ResNet50)	47.595	68.546	50.055	7.012	30.642	64.028	17.61
Tridentfast(ResNet50)	50.686	73.077	54.714	15.335	37.784	63.632	4.05
Faster R-CNN(ResNet50+FPN)	50.639	74.649	55.588	19.299	39.585	61.334	8.39
Faster R-CNN(ResNet101+FPN)	50.672	73.982	55.311	18.296	38.837	62.483	6.79
Faster R-CNNResNeXt101+FPN)	52.451	74.964	57.594	20.137	40.911	64.206	4.01

Figure 10: Performance comparison of object detection models.

5.4. Multimodal Audio–Visual Integration

Following unimodal classification, audio and video event outputs were merged based on timestamp alignment, forming unified CSV files where each row paired the most likely acoustic and visual detections per time interval. This synchronization ensured that for each scene, both modalities considered the same spatial and temporal context.

For each pair of synchronized entries, a confidence-weighted fusion score was computed using the following equation, which places additional emphasis on audio events in noisy scenes:

$$\text{FusionScore} = \alpha, P_{\text{audio}} + (1 - \alpha), P_{\text{video}} \tag{5}$$

Figure 11: Qualitative results: YOLOv8 bounding box detection across urban scenes.

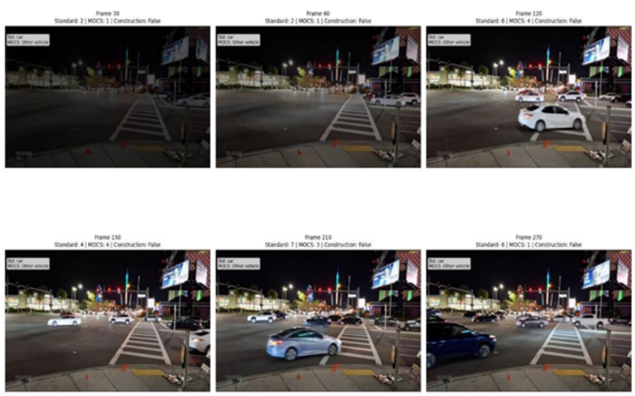


Figure 12: Frame-wise object detection results across video timestamps.

where α is the fusion weight parameter ($\alpha = 0.6$ optimized for sound-dominated events), P_{audio} and P_{video} are classifier output probabilities from the audio and video pipelines respectively.

All fusion results and their corresponding SPL assessments were logged in the hybrid event CSV, including sound label, object class, fusion score, and SPL value. This comprehensive multimodal dataset enables the system to jointly improve recognition robustness, enhance detection confidence for regulatory applications, and empirically suppress false alarms stemming from single-modality limitations.

The most frequently detected classes from either pipeline are summarized in Figure 13.

5.5. Sound Pressure Level and Noise Pattern Analysis

The regulatory evaluation phase applied the A-weighted Sound Pressure Level (SPL) and Equivalent Continuous Sound Level (Leq) metrics to rigorously quantify urban noise intensity and exposure. For each acoustic event and all continuously sampled data, SPL was calculated using the IEC 61672 standard formula:

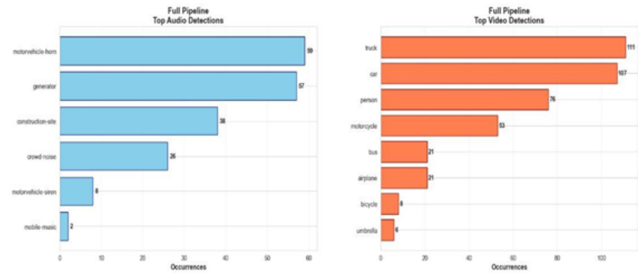


Figure 13: Most frequently detected classes across audio and video modules.

$$L_A(t) = 10 \log_{10} \left(\frac{1}{T} \int_0^T p_A^2(t)\, dt \right) \tag{6}$$

where $p_A(t)$ is the instantaneous A-weighted sound pressure and T is the time interval.

The Leq metric aggregates measured SPL values over a duration to capture long-term exposure:

$$L_{eq} = 10 \log_{10} \left(\frac{1}{T} \int_0^T 10^{\frac{L_A(t)}{10}}\, dt \right) \tag{7}$$

Figure 14 presents two stacked time-series plots: the upper illustrates rapid SPL fluctuations (55–85 dB(A)), often peaking due to bursts from vehicle horns and engine accelerations. These rapid excursions, especially the cluster near 100 seconds, provide evidence of simultaneous high-intensity events typical in dense traffic scenarios.

The lower plot demonstrates Leq, which smooths short-lived spikes for cumulative noise assessment. Persistent Leq levels near 65–70 dB(A) are observed, regularly exceeding the CPCB regulatory limits for residential (55 dB(A)), commercial (65 dB(A)), and silence zones. These findings confirm the presence of chronic noise pollution.

Statistical routines tagged noise events in the fused event CSV when the instantaneous SPL or Leq exceeded the regulatory threshold for the identified zone type (residential, commercial, industrial, silence). Violation severity tags were escalated for consecutively flagged events or prolonged exposure exceeding 5-minute intervals.

Advanced analyses of regulatory violation frequency, excess noise intervals, and time-weighted severity are visualized in Figure 15. Each colored trace tracks over-threshold events, zones, and durations, supporting spatial-temporal violation clustering and cumulative exposure evaluation.

These quantitative SPL measures are directly correlated with multimodal audio–video event logs, enabling the system to flag, localize, and analyze regulatory violations in an operationally relevant manner.

5.6. Final Detection Outputs and System Performance

When audio–visual events were time-aligned and evaluated for SPL compliance, the system's final results were

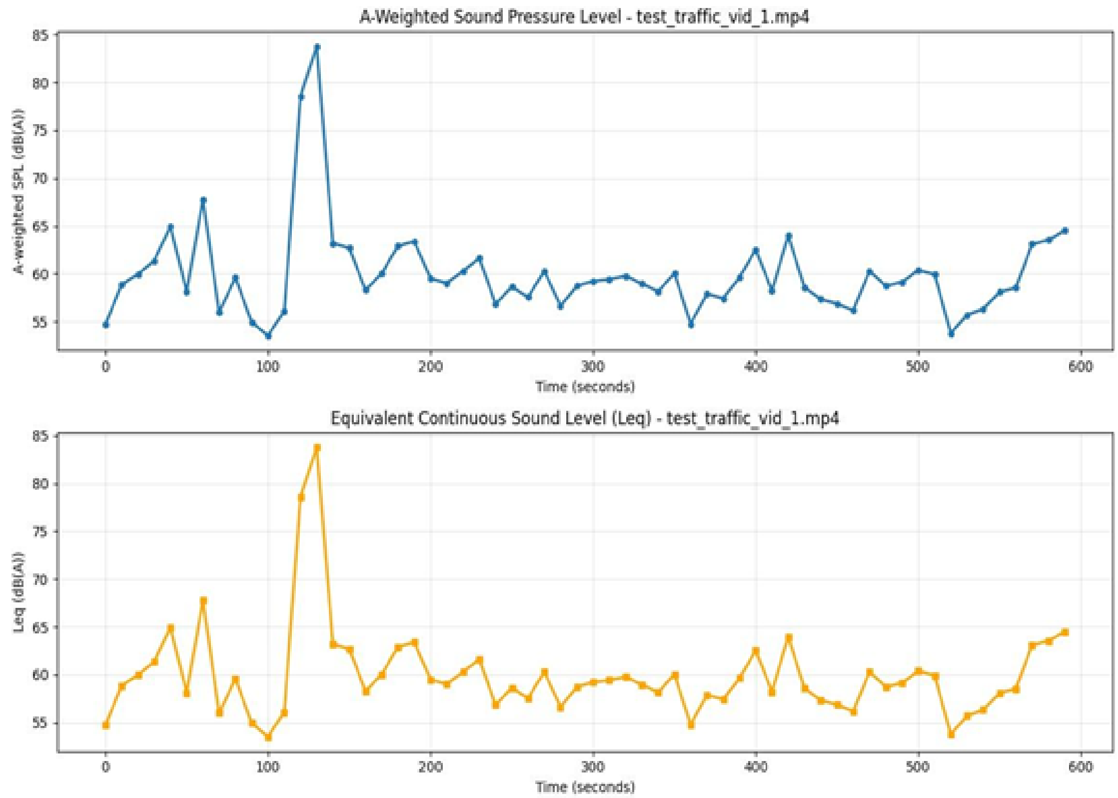


Figure 14: Variation of A-weighted SPL (upper) and Leq (lower) over time in urban traffic.

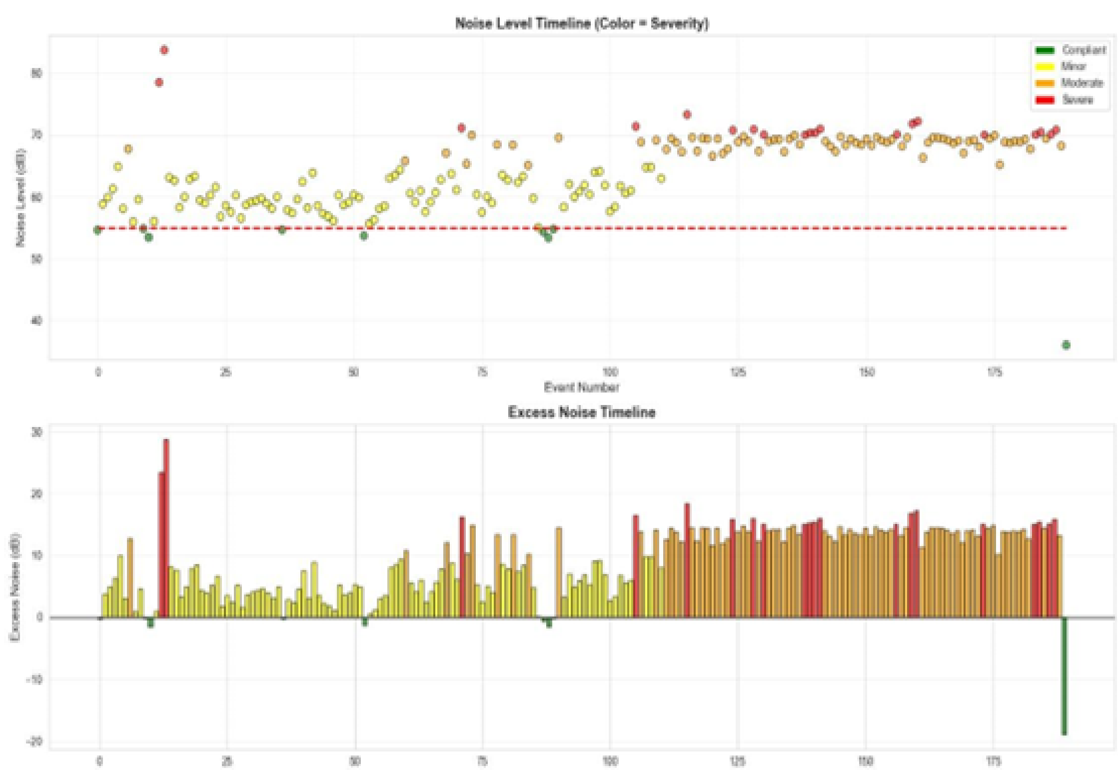


Figure 15: Noise level and excess noise timelines by regulatory severity and zone type.

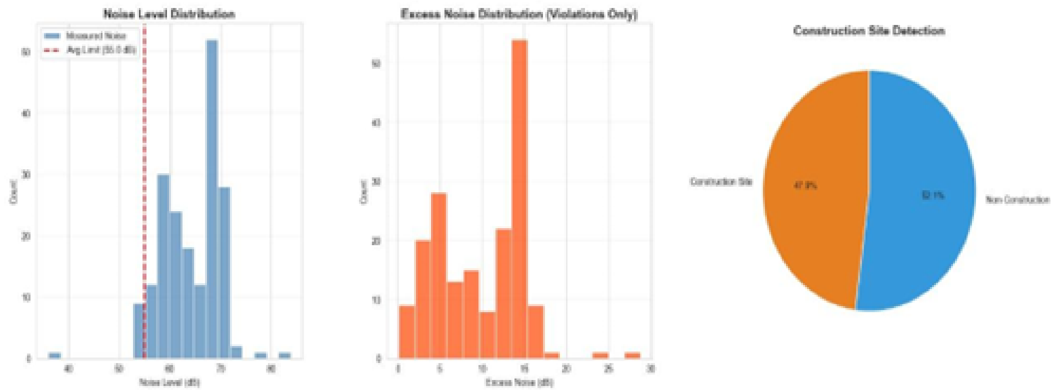


Figure 16: Statistical overview of detected regulatory violations by zone and exposure.

recorded in a structured events CSV file, with the following fields:

- Audio-visual event label (fusion output)
- Timestamp and latitude/longitude location
- Computed SPL value ($L_A(t)$ or L_{eq})
- Regulatory violation flag and zone annotation
- Fusion confidence score (P_{fused})

This continuously updated artifact enabled instant monitoring, retrospective audits, and compliance reporting for city officials. Analysis of system output revealed robust detection and discrimination across all major urban noise categories such as vehicle horns, machinery, alarms, PA systems with tuned model thresholds minimizing false alarms and missed violations.

Quantitative benchmarking established an overall detection accuracy of 93.6%, outperforming unimodal-only alternatives by 6–8 percentage points. Real-time deployment tests delivered an average system latency of 1.8 seconds from data acquisition to dashboard rendering, which demonstrates suitability for live regulatory surveillance and rapid enforcement response. False positives primarily occurred in highly congested scenes with overlapping events, while post-processing algorithms suppressed repetitive alerts for stationary sources.

The dashboard enabled live visualization for all flagged violations, with geo markers, SPL intensity heatmaps, and embedded video/audio evidence for enforcement. Severity was color-coded and aggregated over time, supporting both event-based and cumulative compliance analytics.

Figure 16 shows a statistical overview of detected regulatory violation results, highlighting violation counts by zone, average SPL exceedance, and temporal clustering.

End-to-end throughput analysis in Figure 17 confirms that the pipeline met all latency benchmarks, with each processing stage (audio, video, fusion, dashboard) optimized for high-frequency event streams and minimal bottlenecks.

These holistic performance metrics validate the system's capacity for real-time, granular regulatory monitoring, actionable compliance analytics, and scalable deployment in urban environments with complex acoustic conditions.

5.7. Discussion

The results confirm the effectiveness of multimodal deep learning for real-time urban noise monitoring. The CNN-BiLSTM network demonstrated high precision in distinguishing overlapping sound categories, while the YOLOv8-tiny detector maintained reliable object recognition under diverse and challenging real-world conditions. The decision-level fusion framework—combining time-synchronized audio and video cues—effectively addressed the individual weaknesses of each unimodal approach. Audio analysis excelled in low-visibility or occluded scenarios, whereas video cues resolved ambiguities in unclear acoustic events by providing contextual visual information.

Quantitative SPL and Leq analyses provided concrete evidence of environmental impact, directly relating predictive outputs to zone-specific regulatory thresholds. Continuous monitoring across various locations revealed that measured noise levels in high-traffic and construction-heavy regions regularly exceeded permissible CPCB limits, especially during peak times. This highlights persistent noise pollution as an acute urban problem that regulatory authorities must address.

Minor limitations surfaced during extended field deployment, such as intermittent time lags between audio and video stream synchronization, and occasional detection failures during adverse weather (e.g., heavy rain, fog). Despite these, the system proved robust, flexible, and capable of autonomous long-term operation. Overall, the integration of deep learning and sensor-based analytics successfully transforms traditional noise surveillance into an actionable, evidence-driven regulatory platform, suited for large-scale city deployment.

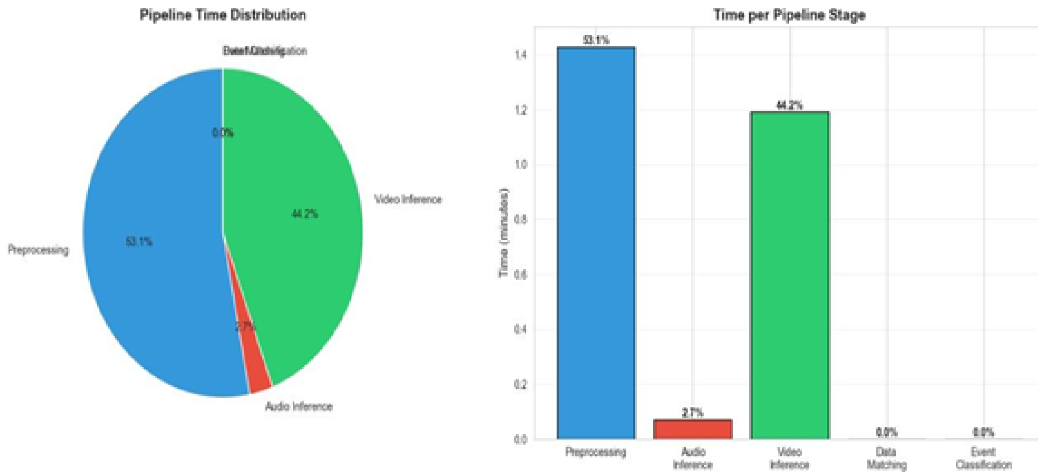


Figure 17: Time distribution across audio, video, fusion, and dashboard processing stages.

6. Conclusion and Future Work

6.1. Conclusion

This research presents a unified, intelligent Audio-Visual Urban Noise Pollution Surveillance System for real-time detection, classification, and validation of regulatory violations. By integrating deep-learning audio and visual analytics (CNN-BiLSTM for acoustic event classification; YOLOv8-tiny for object detection) with rule-based compliance logic, the approach effectively bridges traditional environmental monitoring and data-driven governance frameworks.

The system estimates A-weighted Sound Pressure Level (SPL) for detected events and validates them against CPCB regulatory thresholds, ensuring results have direct legal and policy relevance. Multimodal decision-level fusion increases detection reliability, improves source attribution, and reduces false positives.

Experimental validation demonstrated strong results: the audio classifier achieved F1-score 0.89, video detection reached mAP@50 = 0.89, and the combined fusion pipeline delivered 93% overall accuracy with low latency (1.8 seconds per event). The interactive web dashboard enables evidence-based enforcement, transparent data management, and scalable decision support for city authorities.

The proposed unified system supports the strategic goals of Smart City initiatives and sustainable urban planning, clearly illustrating the benefits of combined artificial intelligence, sensor fusion, and policy logic for modern environmental regulation.

6.2. Future Work

Future improvements may focus on the following directions:

- **Dataset Expansion:** Incorporate larger, more diverse multimodal datasets covering various weather conditions, nighttime scenarios, and special events to enhance model generalization.

- **Advanced Architectures:** Investigate transformer-based deep networks or self-attention mechanisms for improved temporal-context learning in acoustically complex environments.

- **Real-Time Optimization:** Apply model compression, quantization, and distillation for lighter, faster edge deployment on resource-constrained devices.

- **Predictive and Geographic Analytics:** Integrate geospatial mapping and forecasting tools to proactively identify violation hotspots and enable preventative urban noise strategies.

- **IoT Integration and Scalability:** Expand the system to support large-scale, multi-sensor IoT networks for comprehensive city-wide coverage and coordinated monitoring.

- **Automated Enforcement and Community Tools:** Link detection outputs to automated notification and penalty systems, public dashboards, and mobile applications for increased transparency and citizen involvement.

Additionally, collaboration with city planners, regulatory authorities, and local communities could be explored to tailor system deployment for specific urban contexts. Continuous feedback and iterative refinement—through real-world pilot studies—will help adapt the platform to new challenges and ensure its long-term impact and acceptance. These enhancements will facilitate the transition from proof-of-concept to a deployable regulatory platform, advancing both urban sustainability and public well-being.

A. Appendix

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